DICE ROLLER GAME USING PYTHON

Mini Project Report

Project Title: Dice Roller Game Using Python

Programming Language: Python 3

IDE: Visual Studio Code

Author: Add your name here

1. Objective

To develop a simple interactive dice rolling game using Python. The program generates random values for two dice, calculates their total, counts the rolls, and lets the user continue or exit.

2. Abstract

The Dice Roller Game is a beginner-friendly, command-line Python project that simulates rolling two dice. It uses Python's random module to generate values from 1 to 6. The user can roll repeatedly, view each die's value and the total score, and see the number of rolls. The project demonstrates loops, conditional statements, random number generation, and input validation.

3. Problem Statement

Physical dice may not always be available when playing simple games. This program provides a digital alternative that generates and displays random dice values instantly.

4. Software Requirements

• Programming language: Python 3
• IDE: Visual Studio Code
• Operating system: Windows, macOS, or Linux
• Modules: Python standard-library random module

5. Hardware Requirements

• Computer or laptop
• Keyboard and display
• 4 GB RAM recommended

6. Features

• Rolls two dice simultaneously
• Generates random values from 1 to 6
• Calculates the total score
• Counts the number of rolls
• Allows the user to continue or exit
• Handles invalid input

7. Algorithm

1. Start the program.
2. Import the random module.
3. Initialize the roll counter to zero.
4. Ask the user whether to roll the dice.
5. If the user enters 'y', generate two random numbers from 1 to 6.
6. Calculate the total, increase the roll counter, and display the results.
7. If the user enters 'n', display the total roll count and stop.
8. For any other input, display an error message and ask again.

8. Source Code

The complete source code is included in the project ZIP file as **dicerollergame.py**.

9. Sample Output

Example only; random values may differ:

```
===== DICE ROLLER GAME =====
Roll the dice? (y/n): y
Dice 1: 3
Dice 2: 5
Total Score: 8
Roll Count: 1
--------------------------
Roll the dice? (y/n): n
Thanks for playing!
Total Rolls: 1
```

10. Advantages

• Easy to understand and use
• Produces dice values instantly
• Helps beginners practise Python fundamentals
• Can be expanded with more features

11. Future Enhancements

• Add a graphical user interface
• Add dice images and rolling animation
• Store previous roll results
• Add a two-player mode

12. Conclusion

The Dice Roller Game successfully simulates rolling two dice and displays their values, total score, and roll count. It demonstrates essential Python concepts and can be extended into a more advanced game.